

Topic Modeling for Business Reviews

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1 Problem Statement and Objective

Local businesses increasingly rely on their online reputation to attract and retain customers. Online review platforms such as Google Reviews and Yelp provide some of the richest and most accessible sources of authentic customer feedback. However, many independently owned businesses lack the analytical tools needed to extract insights from this unstructured data. Over time, a single business may accumulate hundreds or even thousands of customer reviews covering a wide range of operational themes, including, but not limited to, service quality, cleanliness, wait times, pricing, staff interactions, and product quality. Without a structured analytical framework, valuable patterns within this feedback remain buried in unstructured text, limiting a business owner's ability to make informed, data-driven decisions that improve operations and customer satisfaction.

Current approaches to customer feedback analysis are often manual or rely primarily on aggregate star ratings, both of which overlook the nuanced information contained in written reviews. Star ratings summarize customer sentiment but provide little insight into the specific operational issues driving positive or negative experiences. Consequently, businesses may struggle to identify recurring problems or prioritize areas for improvement.

The objective of this project is to apply natural language processing (NLP) and classical machine learning techniques to Google Reviews collected from a randomly selected, anonymous local business. Specifically, this report analyzes negative customer reviews using topic modeling to identify recurring operational themes that can help business owners prioritize areas of improvement.

2 Methodology

This analysis displays the three modeling approaches utilized during the evolution of our data science strategy to solving the topic modeling problem. In chronological order of implementation and at a high level, the methods used are: K-Means clustering on a TF-IDF document-term matrix, K-Means clustering on SBERT-based semantic embeddings, and Latent Dirichlet Allocation (LDA) topic modeling on a processed review corpus.

These methods were selected to highlight different perspectives on topic discovery and assumptions about review text structure. The TF-IDF-based K-Means model emphasizes lexical patterns, the SBERT-based K-Means model captures semantic similarity beyond exact word overlap, and LDA provides a probabilistic view of latent topics and their distribution across documents. Together, they allow for a qualitative and quantitative comparison of how different modeling choices shape the interpretation of customer feedback.

The following sections describe the data preprocessing, feature extraction, and representation steps required to prepare the reviews for these models, followed by unsupervised clustering and topic modeling results. Figure 1 summarizes the overall analysis and data workflow, including the branching structure from a common cleaned dataset into the three modeling pipelines.

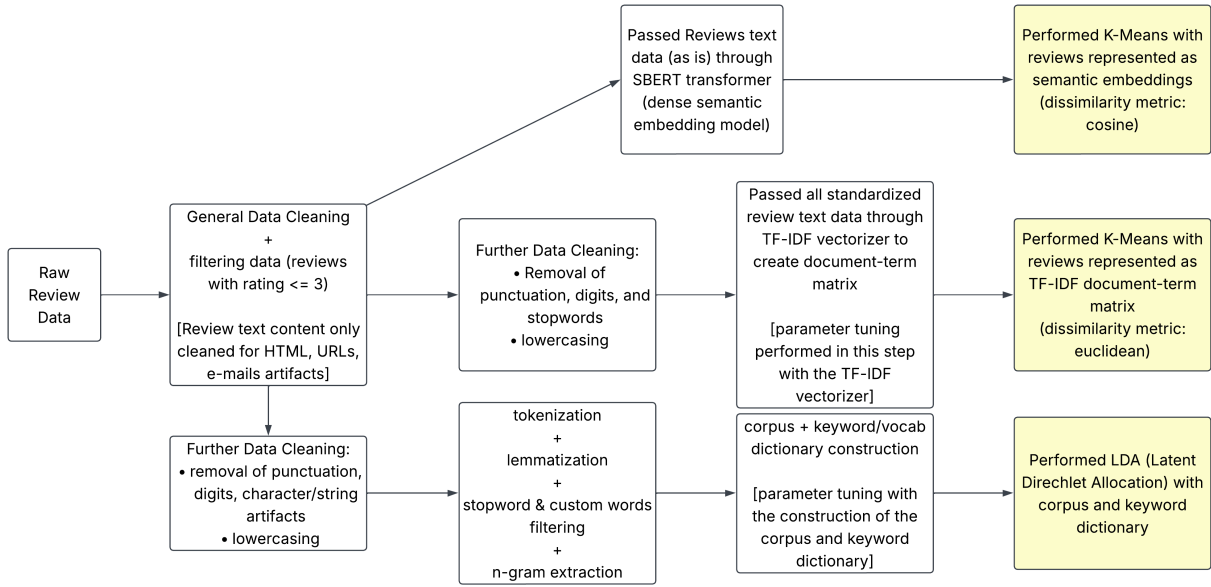


Figure 1: Methodology Pipeline

3 Data Source and Data Collection

All data was collected via Google Reviews from a random and anonymous business using a Google Reviews scraping tool developed by George Khananaev. Google Reviews was selected as the sole data source because it provides an accessible, standardized, and information-rich collection of customer feedback suitable for text analysis. Each extracted review record included the review ID, location ID, reviewer name, review rating, number of review likes, reviewer profile image, reviewer profile URL, review date, review text, and business owner response.

To preserve the anonymity of both the business and its customers, all personally identifiable information—including reviewer names, profile images, profile URLs, and business identifiers—was removed before analysis. The remaining dataset only retains the information necessary for sentiment and topic analysis. Throughout this report, the business is described only at the industry level. Specifically, it operates within the beauty and haircare service industry, providing sufficient context for interpreting the themes identified during topic modeling while protecting the identity of the business.

4 Data Preprocessing, Feature Extraction, and Data Representation

The raw review data first underwent a general cleaning pass in which only the review text was sanitized for HTML and web artifacts, URLs, and e-mail addresses. After this initial cleaning, the dataset was filtered to retain only reviews with ratings less than or equal to 3 out of 5, focusing the analysis on negative customer feedback. From this common filtered dataset, the reviews were routed into three analysis branches, each with preprocessing tailored to the requirements of its respective model.

For K-Means clustering with lexical features, the text data was further standardized by lowercasing and removing punctuation, digits, irrelevant character artifacts, and stopwords. The resulting cleaned review texts were then transformed into a TF-IDF document-term matrix using the `TfidfVectorizer` from `scikit-learn`, with vectorizer parameters later tuned to adjust the TF-IDF matrix and consequently improve clustering performance. For K-Means clustering with semantic features, the reviews—after the initial general cleaning and rating-based filtering—were passed directly into an SBERT transformer to obtain dense

sentence embeddings. No additional text normalization was applied before SBERT, in order to preserve the original wording and stylistic nuances of each review and avoid potential distortion of meaning that could affect the semantic representations.

For LDA topic modeling, the text was processed in a manner similar to the TF-IDF branch but with additional linguistic preprocessing. The reviews were tokenized, lemmatized, and filtered using both standard stopwords and a custom stopword list, and n-grams were extracted to capture meaningful multiword expressions. From this processed corpus, a keyword vocabulary dictionary was constructed and iteratively tuned to support more coherent and interpretable topic keywords.

(Please refer to Figure 1 as this visualizes the data preprocessing, feature extraction, and final data representations pipeline discussed in this section)

5 Unsupervised Clustering for Topic Discovery/Modeling

5.1 K-Means with TF-IDF Matrix

K-Means clustering on a TF-IDF document-term matrix provides a simple baseline for topic discovery in text data. In this approach, each review is represented as a high-dimensional sparse vector whose entries encode the importance of individual terms relative to both the document and the overall corpus. This representation emphasizes distinctive words and phrases that characterize different types of reviews, making it well-suited for grouping documents by lexical similarity.

Using TF-IDF features with K-Means also aligns with classical information retrieval and text mining practice, where bag-of-words models remain effective for capturing *broad topical structure* despite their simplicity. The algorithm partitions reviews into clusters based on distance in the TF-IDF space, and each cluster can then be interpreted by examining the most heavily weighted terms and representative reviews. This combination of a straightforward clustering objective and transparent feature space makes it a useful starting point for topic modeling, against which more complex semantic and probabilistic methods (such as SBERT-based clustering and LDA) can be compared, but it will prove to be an incorrect assumption about the underlying topical structure in review text and as a result produce topical clustering that is difficult to interpret.

Despite the flawed data presupposition in our initial text modeling attempt, the failure yielded valuable insights—specifically, that this text representation method is poorly suited for K-Means clustering. Nevertheless, hyperparameter tuning provided optimal configurations for the TF-IDF vectorizer and the K-Means model (summarized in Figures 2 and 3). Based on these results, we focused on configurations with six or seven clusters. This choice was driven by silhouette scores, which peaked at these cluster counts before showing instability in the marginalized cluster analysis plot. Conversely, the Within-Cluster Sum of Squares (WCSS) metric failed to indicate a clear, meaningful cluster count. Ultimately, the broader tuning analysis suggested that robust clustering was unlikely. This conclusion is supported by Figure 3, where the WCSS curve lacks a distinct “elbow,” and Figure 4, where the silhouette scores remain uniformly low across $k = 2$ to $k = 9$, signaling weak separation between the clusters.

There are several plausible explanations for this behavior. One possibility is that the K-Means algorithm, operating in a sparse TF-IDF space, is not sufficiently expressive to capture the complexity of text data in which multiple topics can coexist within a single review and across the corpus. Another is that the preprocessing applied to the review text may not have been aggressive enough to remove common or domain-specific words that do not contribute to interpretable topical structure, thereby adding noise to the clustering process. There is also a flawed assumption about the model and data representation mentioned previously. Examining the top-25 most heavily weighted terms for the three-cluster K-Means solution (Figure 2) reveals

substantial overlap in keywords across clusters and repeated high-frequency terms, which makes human interpretation of the resulting clusters difficult and further supports the conclusion that this TF-IDF-based K-Means configuration did not yield well-separated, semantically coherent topics.

```
TF-IDF Params: max_features=100, min_df=14, max_df=0.45
K-Mean Params: n_clusters=3, random_state=6740
```

```
-----

Cluster 1:
hair, cut, cut hair, hair cut, stylist
asked, like, lady, haircut, wanted
short, know, want, place, going
guy, worst, horrible, didnt, location
time, uneven, bad, job, experience

Cluster 2:
great, clips, great clips, experience, hair
time, cut, went, location, stylist
going, got, haircut, store, left
said, bad, took, terrible, told
minutes, dont, worst, like, barber

Cluster 3:
haircut, worst, worst haircut, bad, experience
guy, stylist, location, wanted, asked
like, place, good, hair, ive
people, got, time, going, fast
coming, didnt, customers, head, great

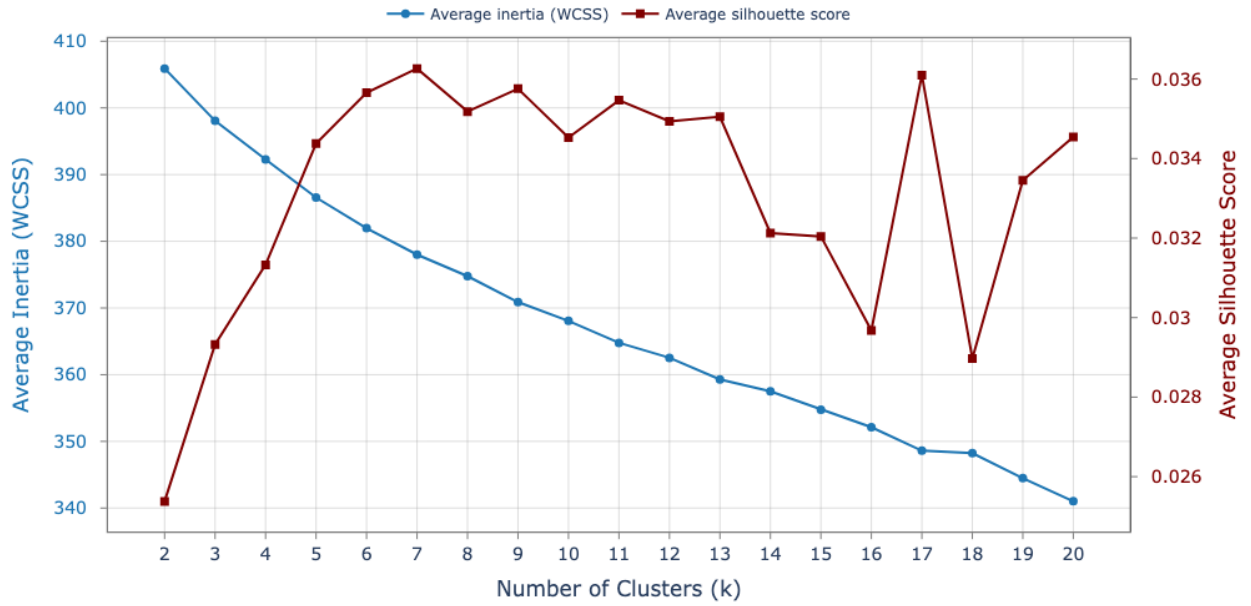
Cluster 4:
service, bad, customer service, customer, haircut
told, asked, place, hair, rude
worst, terrible, time, want, horrible
online, good, cut, check, like
said, barber, experience, recommend, minutes

Cluster 5:
staff, rude, haircuts, haircut, experience
location, customers, bad, listen, worst
saying, place, went, working, customer
salon, dont, unprofessional, left, wait
recommend, got, coming, store, want

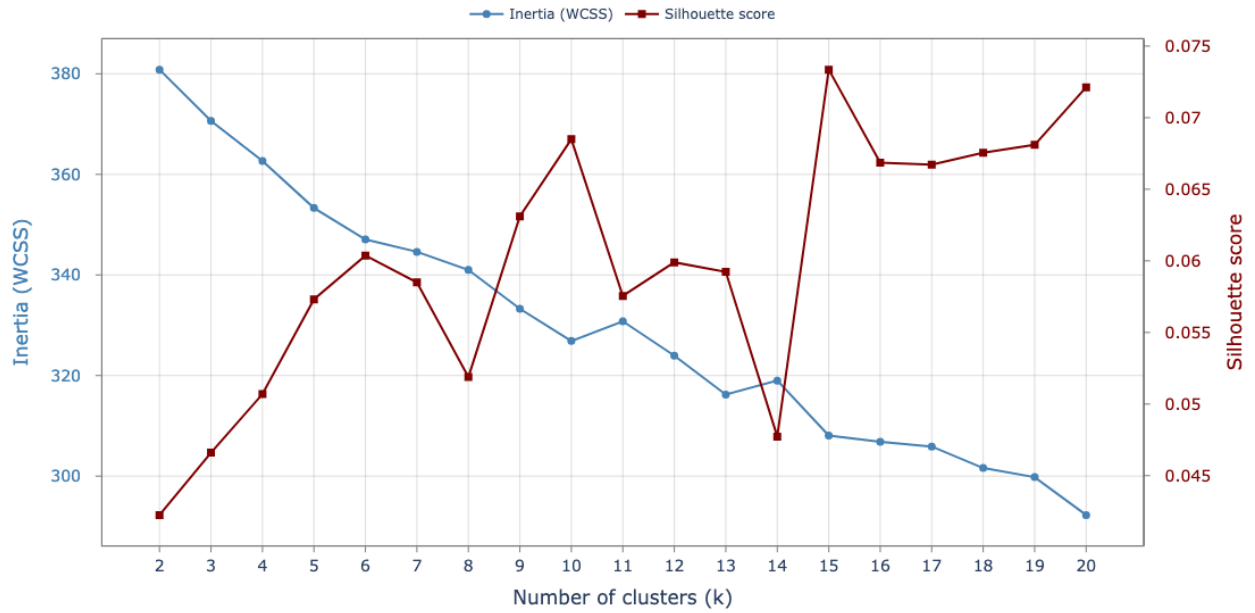
Cluster 6:
wait, place, good, time, haircuts
dont, people, lady, online, minutes
bad, stylist, like, horrible, cut
nice, asked, away, going, experience
close, man, woman, haircut, guy
```

Figure 2: TF-IDF and K-Means Clustering Results (k=6)

Marginalized Cluster Analysis for TF-IDF K-Means: Effect of K-Clusters



K-Means Clustering Analysis using TF-IDF document-term matrix



(TF-IDF vectorizer parameters: max_features=100, min_df=14, max_df=0.45)

Figure 3: TF-IDF based KMeans Clustering Analysis Plots

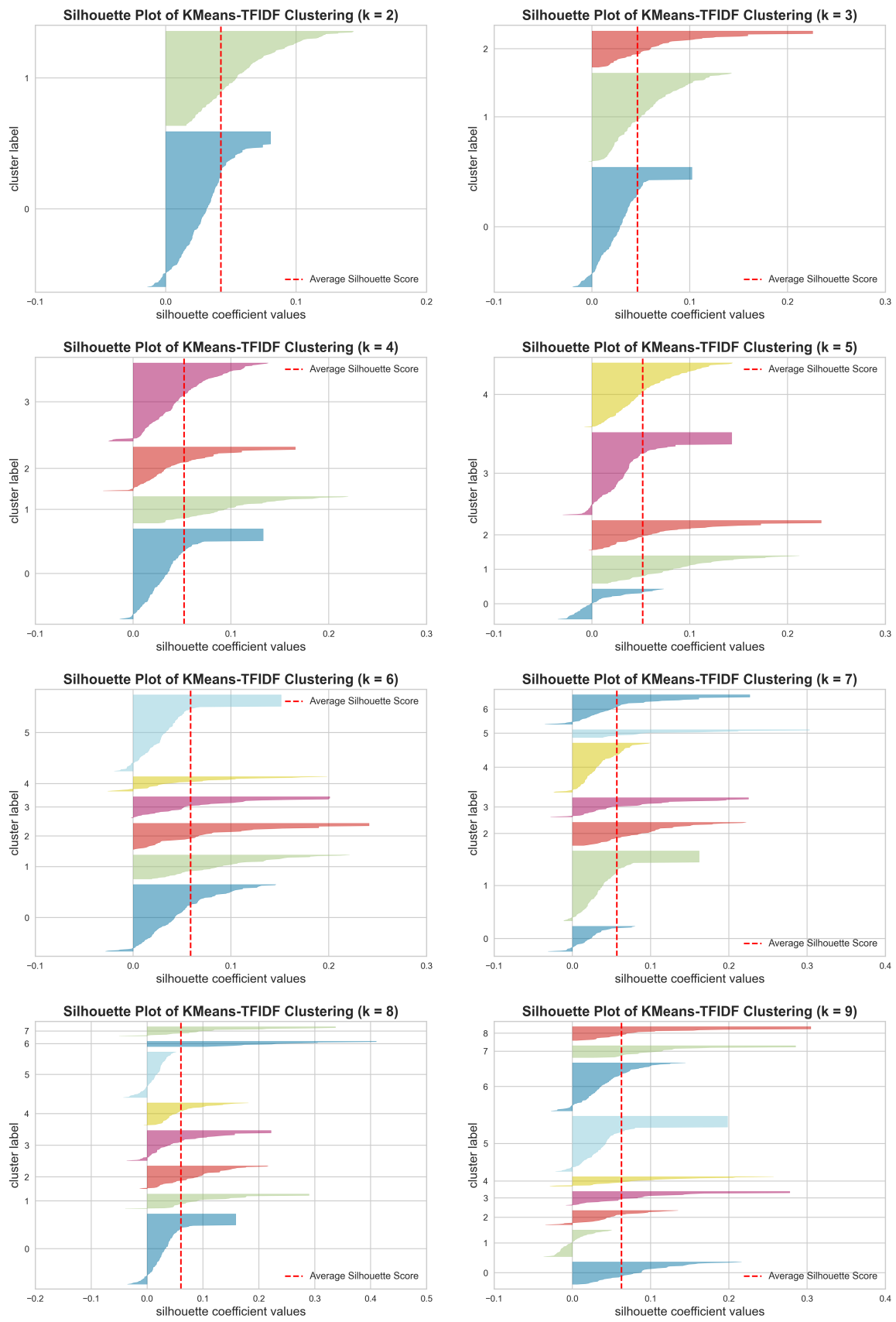


Figure 4: Silhouette Plots for TF-IDF based KMeans

5.2 K-Means with SBERT Semantic Embeddings

K-Means clustering on SBERT semantic embeddings provides a complementary approach to topic discovery that focuses on meaning rather than surface-level word overlap. In this method, each review is encoded as a dense vector using a pre-trained SBERT transformer, capturing contextual relationships among words and phrases within the review. As a result, reviews that express similar sentiments or discuss similar themes can be mapped close together in the embedding space even when they use different vocabulary or phrasing.

Applying K-Means in this semantic space allows clusters to form based on cosine similarity between embeddings, emphasizing conceptual similarity rather than exact term frequencies. This is an improvement in text data representation and makes the SBERT-based approach particularly suitable for business reviews, where customers may describe comparable experiences in varied language, and where topics often span multiple expressions of the same underlying issue. On the other hand, this method of transforming the data and hard clustering still assumes a single-topic structure in the text data, which is problematic since our task is to discover operational themes found in the review data. But even with the limitations of this clustering method, we can still attempt to infer topics from the resulting clusters. In the context of this project, the SBERT clustering results serve both as a richer semantic counterpart to the TF-IDF-based clusters and as a bridge between purely lexical models and probabilistic topic models such as LDA.

A key modeling choice in this branch is the use of cosine dissimilarity when clustering SBERT embeddings with K-Means. Mathematically, K-Means is defined to minimize the within-cluster sum of squared Euclidean distances between points and their cluster centroids. However, for vectors that are ℓ_2 -normalized to unit length, Euclidean distance and cosine similarity are monotonically related: for two unit vectors $\mathbf{x}$ and $\mathbf{y}$,

$$\|\mathbf{x} - \mathbf{y}\|_2^2 = \|\mathbf{x}\|_2^2 + \|\mathbf{y}\|_2^2 - 2\mathbf{x}^\top \mathbf{y} = 2 - 2\cos(\theta),$$

where $\cos(\theta) = \frac{\mathbf{x}^\top \mathbf{y}}{\|\mathbf{x}\|_2 \|\mathbf{y}\|_2}$ is the cosine similarity between $\mathbf{x}$ and $\mathbf{y}$. Thus, minimizing squared Euclidean distance between unit-normalized embeddings is equivalent to maximizing cosine similarity, and K-Means remains mathematically well-grounded as a clustering objective in this normalized embedding space while effectively operating on cosine-based semantic similarity.

From the clustering analysis summarized in Figure 6, we observe a more pronounced bend in the elbow plot compared to the TF-IDF case, but the silhouette scores drop sharply for $k > 3$. This suggests that, as with K-Means on the TF-IDF document-term matrix, the underlying text data remain difficult to separate cleanly into well-defined clusters. The grid of silhouette plots in Figure 7 reinforces this conclusion: for most choices of k , a large proportion of points exhibit negative silhouette values, indicating that many reviews lie closer to a different cluster than the one to which they were assigned.

Despite these quantitative indicators of weak cluster structure, the semantic clusters produced by K-Means on SBERT embeddings still yield improved groupings with better interpretability at small k -values. Focusing on the $k = 2$ solution in Figure 5, we find that cluster 0 predominantly groups reviews that describe unprofessional or inappropriate conduct by staff, whereas cluster 1 contains reviews that emphasize the quality of service. In other words, even though the elbow and silhouette diagnostics point to limited separation in the embedding space, the resulting clusters remain semantically meaningful and provide actionable insights—an improvement over the TF-IDF-based K-Means results, where lexical overlap made human interpretation of clusters substantially more difficult.

```

=====
CLUSTER 0 | 133 reviews (top-3 representative reviews)
=====

--- Representative #1 | Cosine Similarity to Centroid: 0.7774 ---

Review:
Rude employees, very bad experience!

--- Representative #2 | Cosine Similarity to Centroid: 0.7580 ---

Review:
Very bad service I had to walk out

--- Representative #3 | Cosine Similarity to Centroid: 0.7350 ---

Review:
Well I checked in online reached there on time yet this guy with bad sense
of humor cut other customers first who were behind in queue. He was a thin
guy of asian ethnicity not sure what his name was and he was so
unprofessional. Making stupid jokes of all kind he had hard time
pronouncing my name and called me bahubali said this is easier to call.
What a dumb thing to say. Well i liked there an elderly lady she does a
nice cut but because of this gentleman I'm not going there again.

=====
CLUSTER 1 | 329 reviews (top-3 representative reviews)
=====

--- Representative #1 | Cosine Similarity to Centroid: 0.9107 ---

Review:
My husband and I went in for haircuts this past weekend and I wish we would
have waited and gone to our usual location. The one star I'm giving is
mainly due to the friendliness of the staff; that I won't deny, as for the
quality of our haircuts I can't say the same, we were 100% dissatisfied
with the outcome. My poor husband has decided to wear a baseball cap until
his grows back, and my hair is totally uneven. We felt that once they began
working on us, they were just in a rush to get us out of the seat and onto
the next customer. We were deeply disappointed that we basically paid
someone to destroy our hair. Unfortunately, we will never return to this
location again.

--- Representative #2 | Cosine Similarity to Centroid: 0.8951 ---

Review:
Worst experience everrrr...!!! This branch of X sucksss...Me and a
friend of mine went for a hair cut, stylist chose to give service to my
friend alone and she was not even nice while talking to him. She looked so
upset and showed so much attitude when my friend asked her to trim little
more. She said I don't cut twice. Like seriously? On the other hand, she
did not give service to me saying she has hand pain and asked me visit some
other day. Why would I visit your place again after such extremely horrible
treatment of customers and service. "WORST STYLIST"

--- Representative #3 | Cosine Similarity to Centroid: 0.8909 ---

Review:
It was the worst haircut experience ever. The barber was ready to listen to
what style I want. The staff was very rude. Will not visit this location
again ever.

```

Figure 5: K-Means with SBERT Embedding Results (k=2)

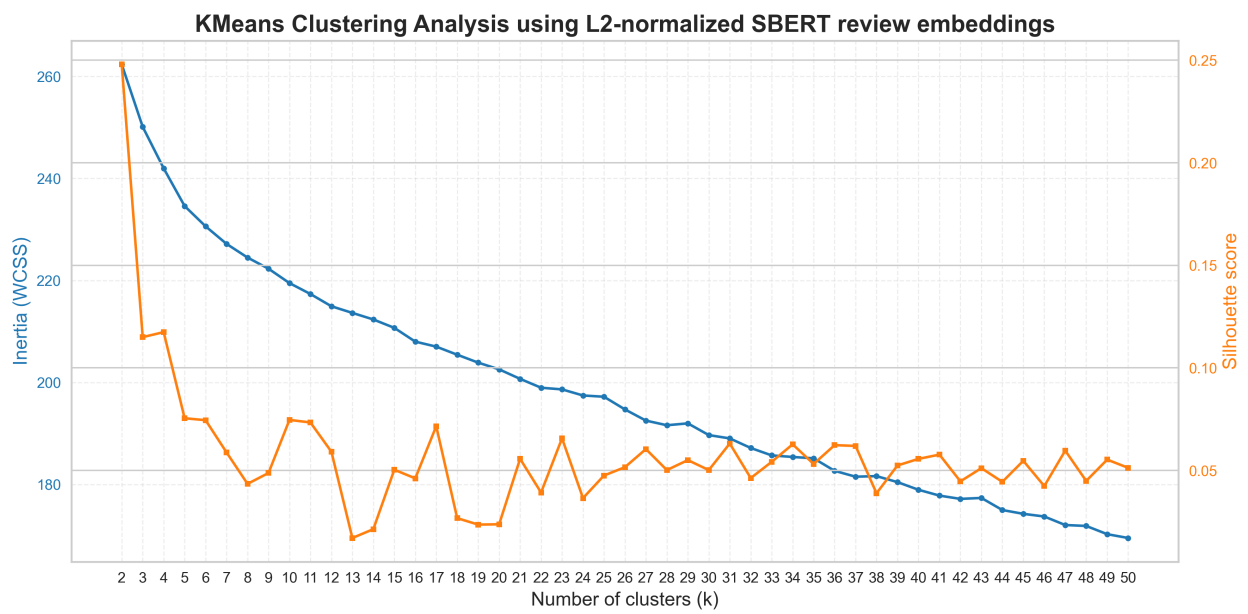


Figure 6: SBERT-based KMeans Clustering Analysis Plot

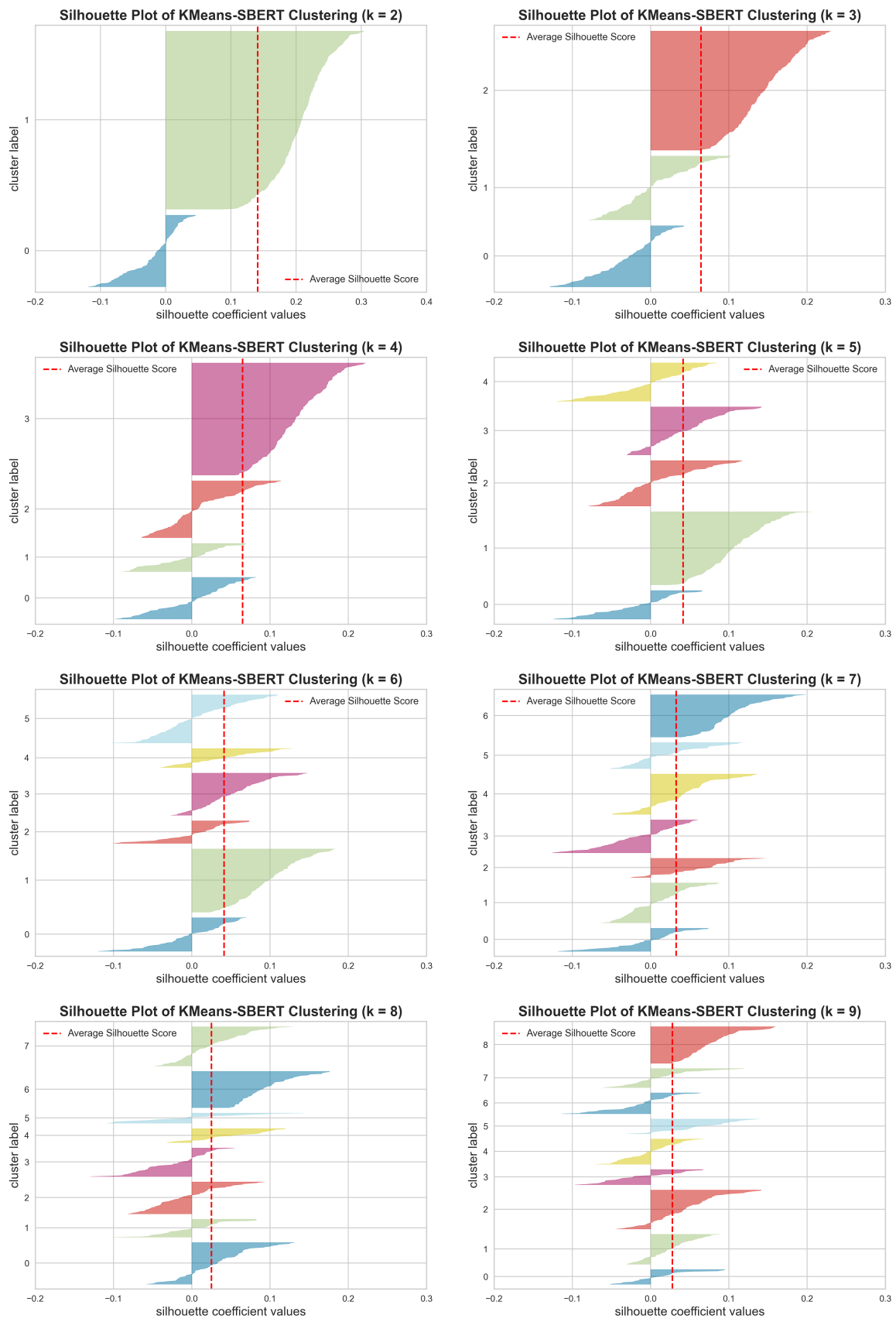


Figure 7: Silhouette Plots for SBERT-based KMeans

5.3 Latent Dirichlet Allocation (LDA)

Latent Dirichlet Allocation provides a probabilistic framework for topic modeling that explicitly represents each review as a mixture of latent topics and each topic as a distribution over words. In this approach, the preprocessed review corpus is converted into a bag-of-words representation using the constructed vocabulary dictionary, and LDA infers the underlying topics by estimating the word distributions and topic proportions that best explain the observed text. This generative view of the data allows individual reviews to express multiple topics simultaneously, which is particularly important for business reviews that often mention several aspects of an experience within a single document.

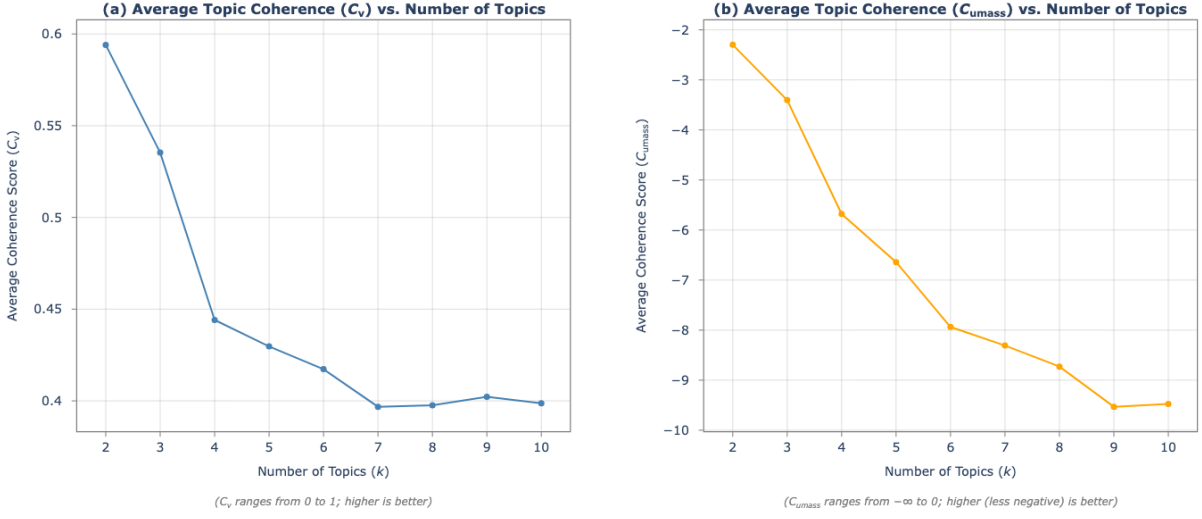
Conceptually, LDA complements the clustering-based methods by shifting focus from partitioning reviews into discrete groups to modeling the continuous structure of topics across the corpus. Whereas K-Means assigns each review to a single cluster, LDA yields topic mixtures that capture nuanced overlaps, such as reviews that discuss both staff behavior and service quality. The resulting topic-word distributions and document-topic weights provide interpretable building blocks for summarizing recurring themes, comparing the prevalence of issues, and identifying which types of complaints are most characteristic of negative reviews. This makes LDA a natural choice for the core topic modeling component of the project and a useful reference point for evaluating the lexical and semantic clusters obtained from the K-Means analyses.

After tuning the vocabulary filtering and text preprocessing steps to obtain an LDA model with the most human-interpretable topics, we found that a good range for the number of topics was two to three based on quantitative performance metrics, and ultimately selected a three-topic model. This choice was broadly consistent with the K-Means analyses, which also suggested a small number of dominant groupings. Using the dictionary filtering parameters summarized in Figure 8, we obtained exceptionally high topic coherence scores for both C_v and u_{mass} , indicating strong semantic similarity among the top keywords within each topic and suggesting that the topics form interpretable thematic structures.

Inspection of the top keywords for the three topics (Figures 9, 10, and 11) confirms this interpretation. Topic 1 centers on negative customer experiences involving service mistakes, uneven or unsatisfactory haircuts, and feelings of disrespect or frustration, often leading customers to leave early, avoid returning, or discourage others from visiting the salon. Topic 2 focuses on complaints related to wait times, perceived poor customer service, and the overall in-salon process (waiting, being attended to, and feeling uncared for). Topic 3 captures issues surrounding unprofessional barber behavior and communication style that made customers feel uncomfortable, disrespected, or annoyed throughout their appointments. Together, these topics provide a coherent summary of the main operational concerns driving negative reviews with clear topic separation as shown in Figures 9, 10, and 11.

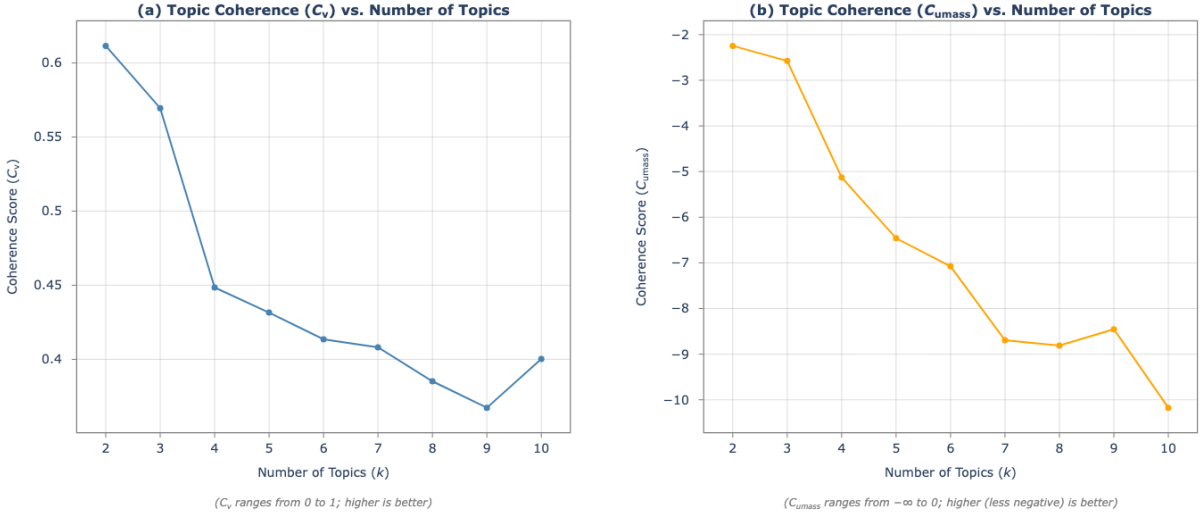
Notably, some of the themes identified by LDA align with the semantic clusters obtained from K-Means on SBERT embeddings: issues around staff conduct, service quality, and the in-salon experience emerge in both models. This convergence between probabilistic topic modeling and embedding-based clustering strengthens confidence that these themes represent stable, underlying patterns in the review corpus rather than artifacts of a specific modeling choice.

Marginalized Topic Coherence Analysis: Main Effect of Topic Count



(Average C_v and C_{umass} coherence scores per topic count. Values are averaged across all corpus-dictionary construction and LDA algorithm settings to show the macro-trend of topic size performance)

LDA Topic Coherence Analysis For Corpus-Dictionary Filtering Preset



(Corpus-Dictionary filtering parameters: $no_below=3$, $no_above=0.40$)

Figure 8: LDA Topic Coherence Analysis Plots

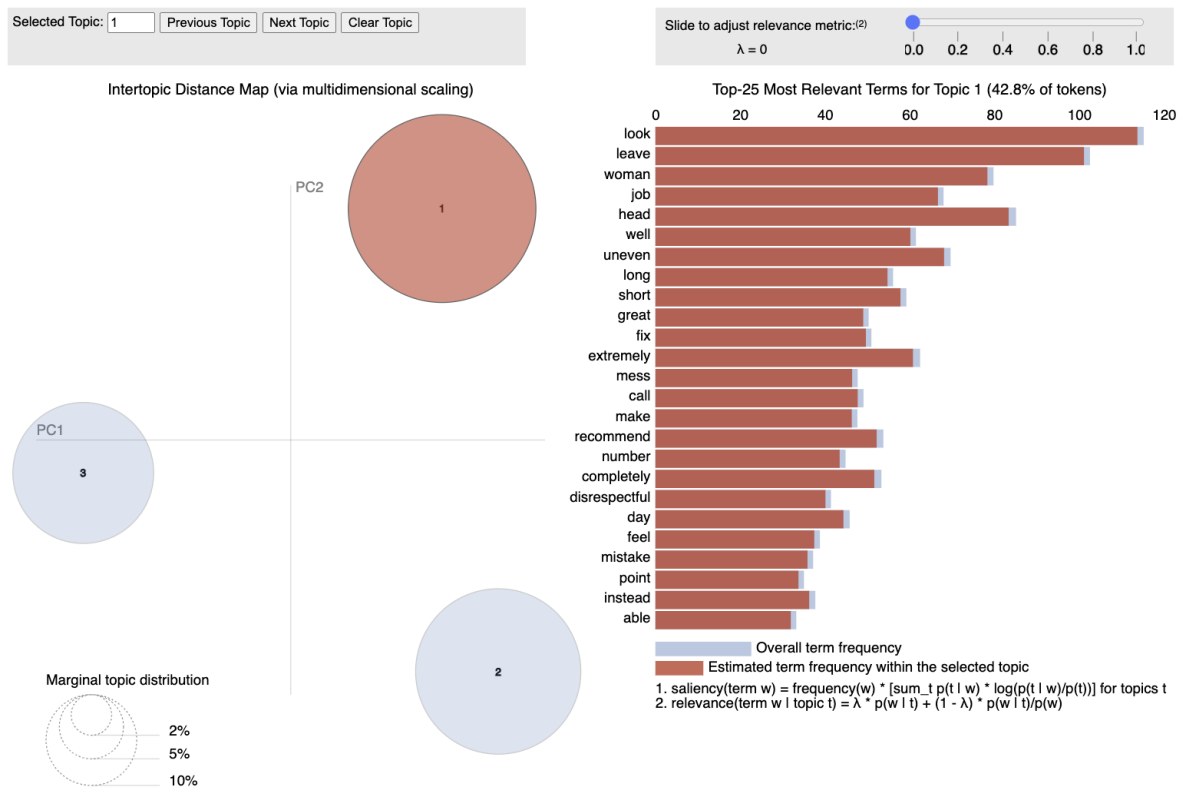


Figure 9: LDA topic 1 Visualization

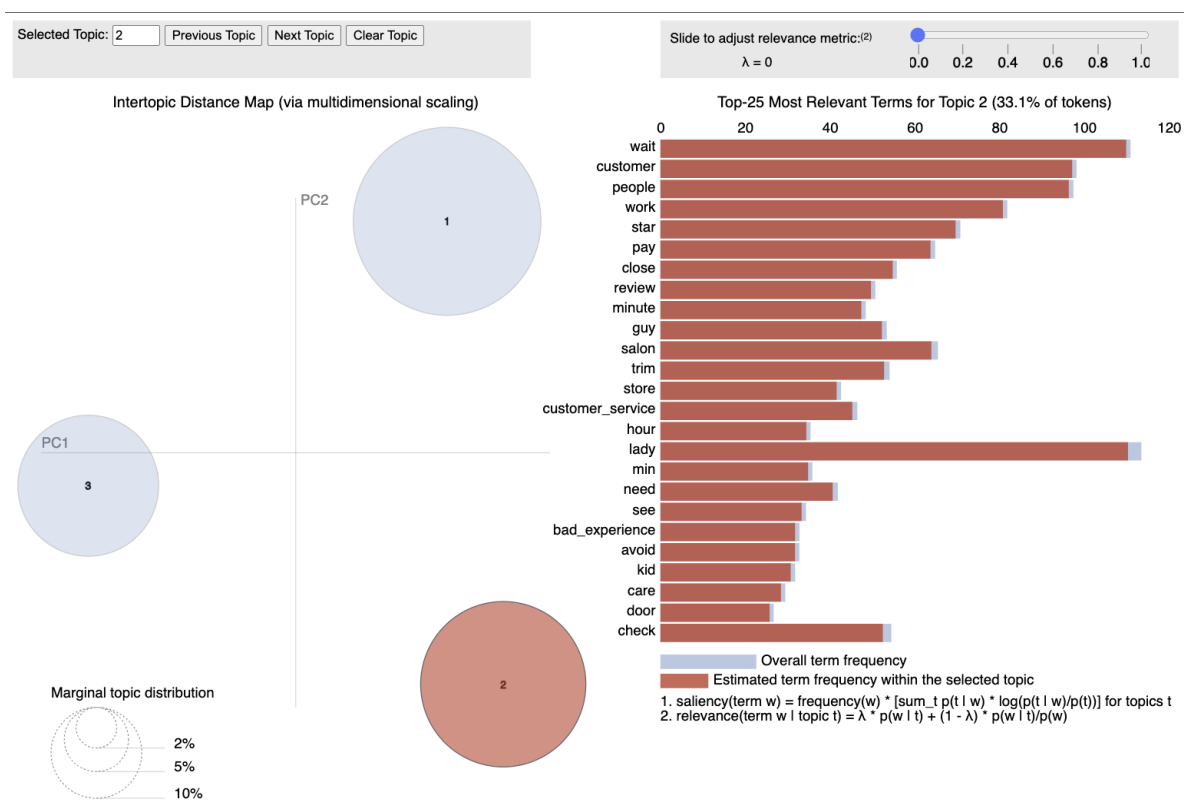


Figure 10: LDA topic 2 Visualization

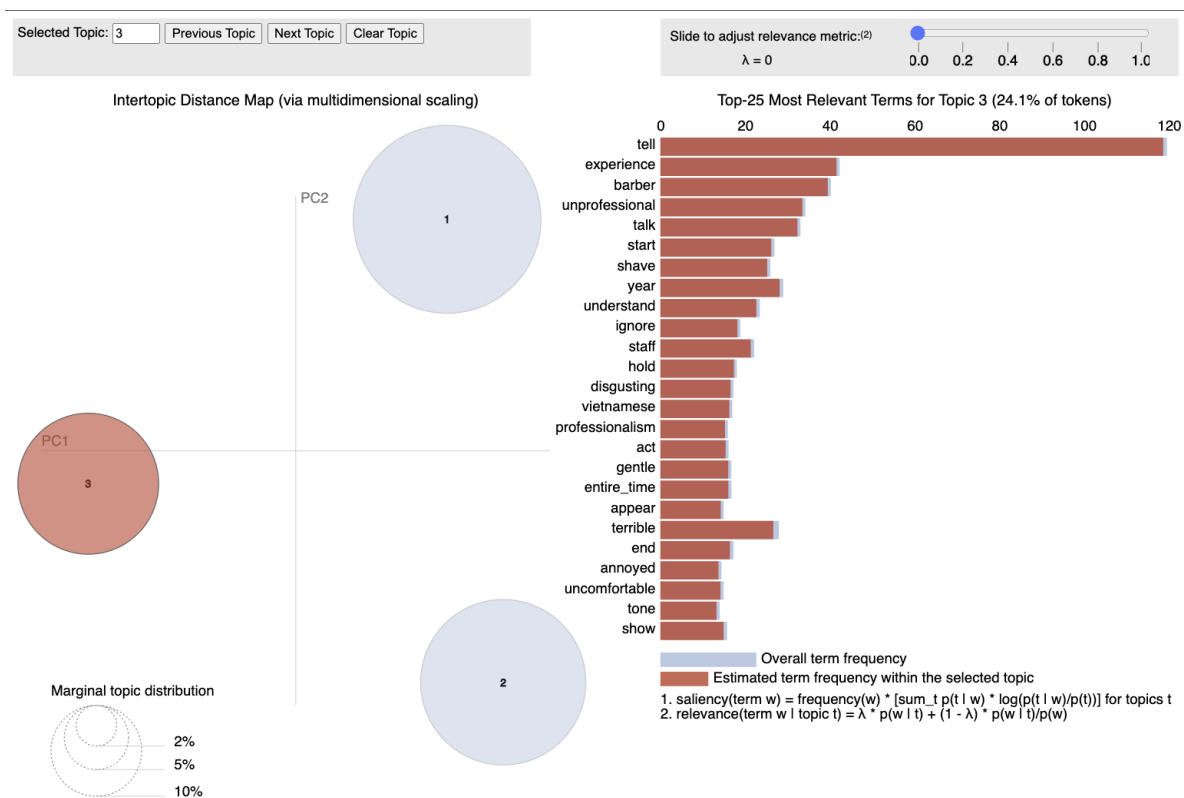


Figure 11: LDA topic 3 Visualization

6 Results and Conclusion

This analysis showed that unsupervised text analysis can reveal meaningful structure in negative customer reviews, while also highlighting the inherent difficulty of cleanly partitioning such text into discrete groups. Across both K-Means approaches, the clustering results suggested poor separation, indicating that customer complaints often span multiple topics and do not naturally fall into sharply defined clusters. Even so, the SBERT-based K-Means model produced more semantically interpretable groupings that distinguished reviews focused on unprofessional staff conduct from those centered on service quality issues, underscoring the advantage of semantic representations over purely lexical features.

LDA provided the clearest and most coherent thematic summary of the corpus. After tuning the preprocessing pipeline and vocabulary construction, the three-topic model yielded stable, interpretable themes related to service mistakes and haircut quality, wait times and in-salon experience, and unprofessional or disrespectful staff behavior. These topics aligned closely with the clusters observed in the SBERT-based K-Means analysis, suggesting that the main patterns identified are robust across different modeling frameworks.

This analysis demonstrates that data science outcomes are heavily dictated by the alignment between methodology and the inherent structure of text data. Initially, applying a single-topic-per-document assumption via TF-IDF and SBERT K-Means hard clustering failed because customer reviews naturally contain multiple themes, resulting in poor topic coherence and low interpretability. Shifting to a multi-topic-per-document assumption through Latent Dirichlet Allocation (LDA) soft clustering provided a major technical breakthrough, yielding significantly higher coherence and clear semantic separation. Ultimately, this methodological evolution revealed that negative reviews for the business are dominated by a small set of recurring operational issues—specifically poor service quality, unpleasant staff interactions, and a frustrating appointment process. For the business owner, these findings provide a highly interpretable roadmap for impactful interventions, proving that data preparation and model selection are only as effective as the underlying assumptions made about the data.

Future work could extend this analysis by incorporating domain-specific preprocessing, exploring alternative topic models, or contrasting themes in positive versus negative reviews to identify what drives satisfaction as well as dissatisfaction.